

UAS Certification

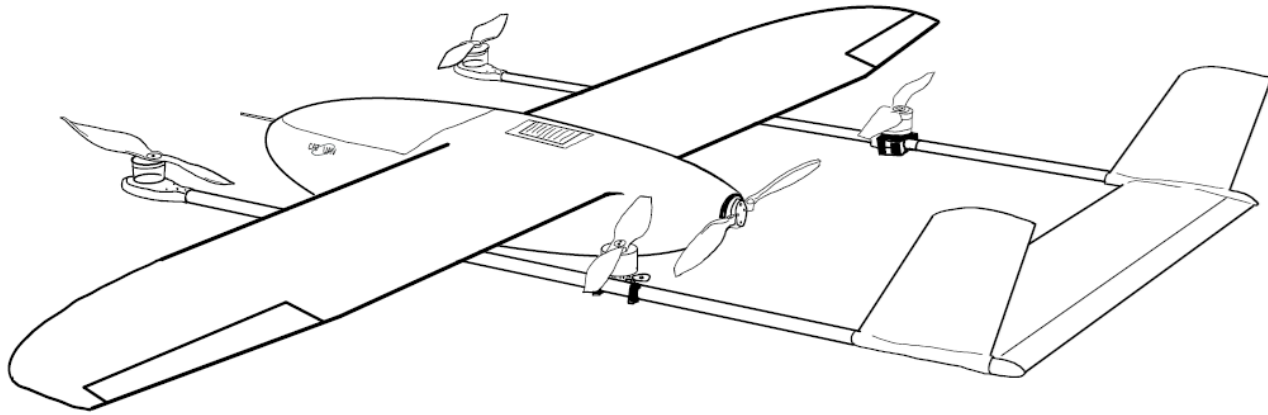
Design and Construction of Drone

Introduction to European Certification Framework

Training by BCN Drone center hosted by Birdview Mapping



SORA Methodology



Scale Operations

SAIL I-II

Operator Responsibility

SAIL III

Compliance Declaration

SAIL IV

Design Verification

SAIL V-VI

Type Certificate

For BIRDVIEW MAPPING
A. H. H. H.
Proprietor

EXECUTIVE COST SUMMARY

System / Program	Amount (₹)	Description
VTOL UAV System	₹39,000,000	Custom VTOL design, engineering, integration, payload, communications, manufacturing and testing
Multicopter UAV System	₹26,500,000	Autonomous multicopter design, LiDAR/RGB payload integration, manufacturing and testing
Common / Program	₹18,000,000	Testing, R&D, training, documentation, project management and support
Additional Logistics Cost	Rs 1,500,000	Travel, Visa, transportation cost and stay for the Spain team for 1 month cost
TOTAL	Rs 85,000,000	

For BIRDVIEW MAPPING

 Proprietor

EU UAS Certification Overview



Certification of the 3 UAS operational elements



UAS Technology (system of systems)



Pilots



Operator (entity with multiple pilots: company, university, administration...)

Certification Bullet Points



Certification requirements are based on the Risk Level



Evaluation is based on Performance



Mandatory Third Parties: Notified Body, NAAs and EASA

EU UAS Regulation Overview



UAS Operation Categories



Open Category

Low Risk



Specific Category

Medium Risk



Certified Category

Traditional Aviation Risk



EU UAS Regulation Overview



UAS Certification Requirements



Open Category

Low Risk

Class Identification Label (C0 to C4)
+ CE Marking

Specific Category

Medium Risk

Option 1:
Class Identification Label (C5 to C6)
+ CE Marking

Option 2:
Operational Authorisation + CE Marking

Certified Category

Traditional Aviation Risk

Type Certificate (TC)



Open Category Certification

Regulation Requirements for Open Category



Open Category

Low Risk

Toys & Mass Market Drones
VLOS Professional Apps



Requirements

CE Marking:

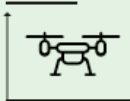
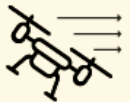
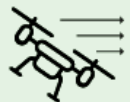
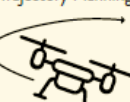
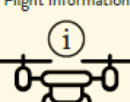
- CIL: Class Identification Label (C0 to C4)
- Radio Equipment Directive 2014/53
- Electromagnetic Compatibility 2014/30
- Machinery Directive 2006/42

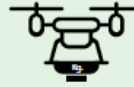
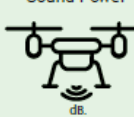
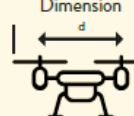
How we can help you

1. Legal Consultancy on selecting Class
2. Pre-Evaluation Testing
3. Evaluation Testing
4. Conformity Assessment

CIL Evaluation Tests



Outdoor Test	PERFORMANCE	C0	C1	C2	C3	C4	C5	C6
	Maximum Height 	✓	✓	✓	✓	✗	✗	✗
	Maximum Flight Speed 	✓	✓	✗	✗	✗	✗	✓
	Low Speed Mode 	✗	✗	✓	✗	✗	✓	✗
	Trajectory Planning 	✗	✗	✗	✗	✗	✗	✓
	Safely Controllable 	✓	✓	✓	✓	✓	✓	✓
	Flight Information 	✗	✗	✗	✗	✗	✓	✓
	Follow-me Mode 	✓	✓	✗	✗	✗	✗	✗

Indoor Test	UAS	C0	C1	C2	C3	C4	C5	C6
	MTOM 	✓	✓	✓	✓	✓	✓	✓
	Electric Powered 	✓	✓	✓	✓	✗	✓	✓
	Sound Power 	✗	✓	✓	✓	✗	✓	✓
	Max. Characteristics Dimension 	✗	✗	✗	✓	✓	✓	✓
	Mechanical Strength 	✓	✓	✓	✗	✗	✗	✗

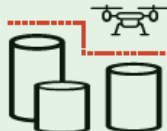
CIL Evaluation Tests

Indoor Test

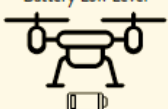
DOCUMENTATION	C0	C1	C2	C3	C4	C5	C6
User Manual 	✓	✓	✓	✓	✓	✓	✓
Information Notice  (EU)	✓	✓	✓	✓	✓	✓	✓

Indoor Test

Outdoor Test

FLIGHT CONTROL	C0	C1	C2	C3	C4	C5	C6
Unique Physical Serial Number 	✗	✓	✓	✓	✓	✓	✓
Remote ID (flight) 	✗	✓	✓	✓	✓	✓	✓
Geoawareness 	✗	✓	✓	✓	✗	✗	✗
Airspace Limitation Function 	✗	✓	✓	✓	✗	✓	✓

Outdoor Test

SAFETY & SECURITY	C0	C1	C2	C3	C4	C5	C6
Loss of C2 Link 	✗	✓	✓	✓	✗	✓	✓
Maximum Flight Speed 	✗	✓	✓	✓	✗	✗	✗
Lights (Flight) 	✗	✗	✓	✗	✗	✓	✗
Battery Low Level 	✗	✓	✓	✓	✓	✓	✓
C2 Links Protections 	✗	✗	✗	✓	✗	✓	✓
Flight Termination System 	✗	✗	✗	✗	✗	✓	✓
Tethered UAS 	✗	✗	✓	✓	✗	✗	✗

Specific Category Certification

Regulation Requirements for Specific Category



Specific Category

Operation Depending Risk

Most UAS Professional Apps:
BVLOS operation, Drone Swarms, Delivery,
Populated Areas...



Requirements

Option 1:

Class Identification Label (C5 to C6)

+

CE Marking

Option 2:

Operational Authorisation (OA)

+

CE marking

How we can help you

1. Legal Consultancy: Best Scenario
2. Pre-Evaluation Testing
3. Evaluation Testing
4. Conformity Assessment

1. Safety Study (SORA, any SAIL)
2. Operational Authorisation Processing
3. Experimental Flights Authorisation
4. EASA Design Verification Process

Operational Authorisation Process

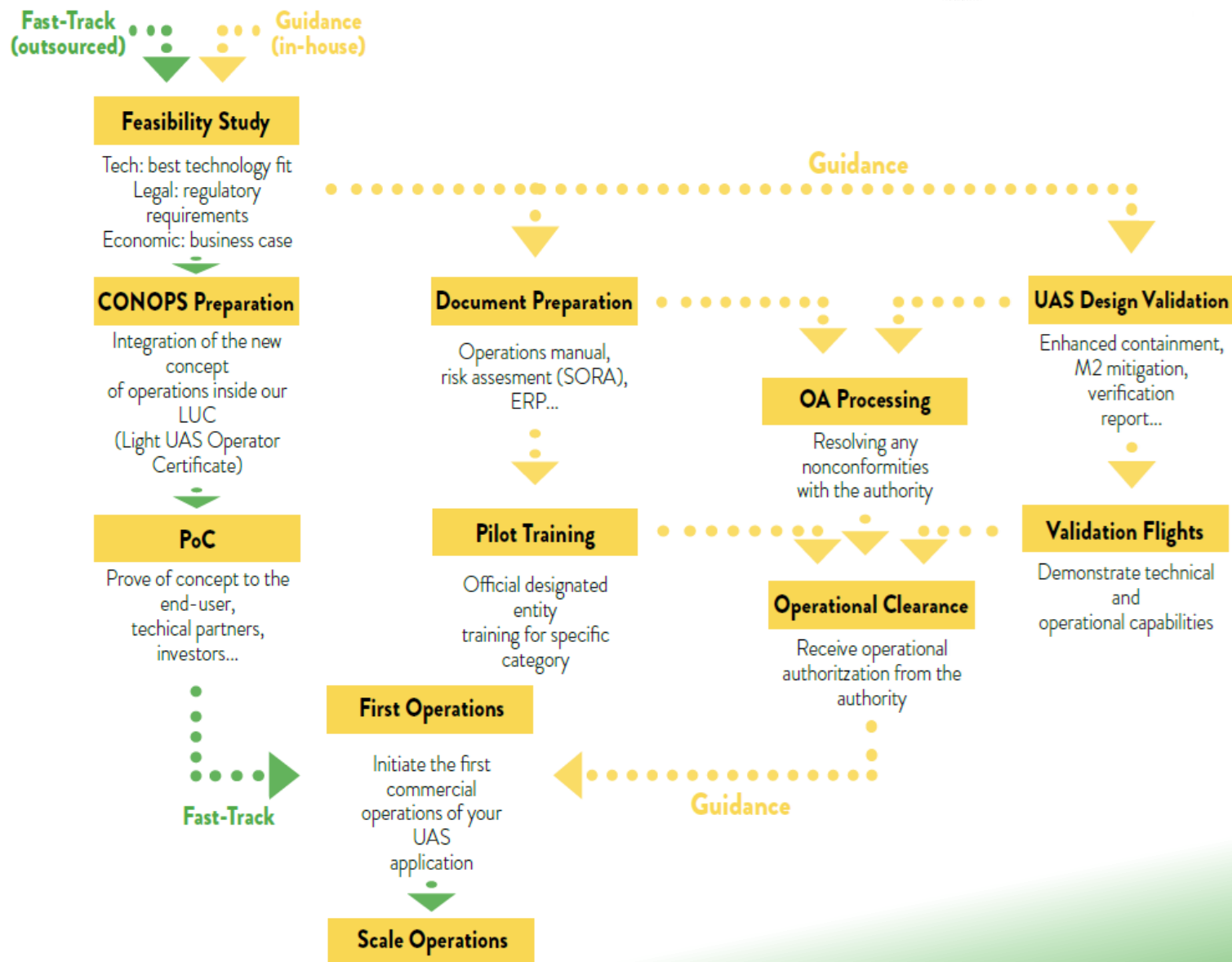


End-to-End Solution

We operate using our LUC privileges to showcase your application to the end-user

Guidance Support

We empower you as operator through flight permits, pilot training, and validation flight tests



Certified Category Certification



Regulation Requirements for Certified Category



Certified Category

Traditional Aviation Risk

Transport of People & Dangerous Goods
Large UAS over populated areas



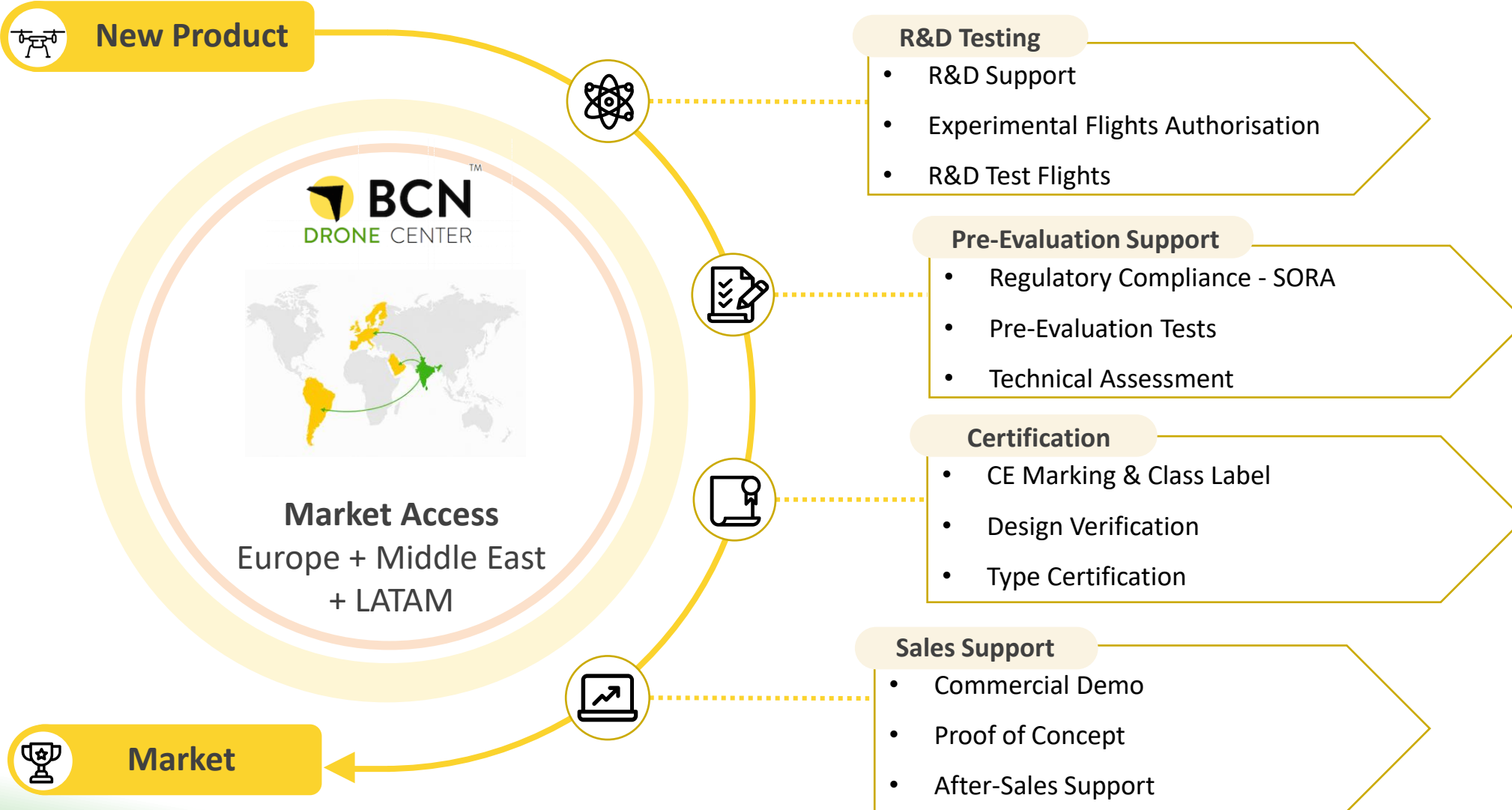
Requirements

Type Certification
Design Organization Approval (DOA)
Production Organization Approval (POA)

How we can help you

1. Legal Consultancy: Best Scenario
2. Experimental Flights Authorisation
3. Pre-Evaluation Testing
4. Evaluation Testing
5. Type Certification
6. DOA Processing
7. POA Processing

One Stop Shop Market Entry Support





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